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Web Technology

QualityLaboratoryManual

ExperimentNo. 05

Program to demonstrate concept of DTD and its types



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ExperimentNo.05

Title of Experiment: Program to demonstrate concept of DTD and its types

Aim of Experiment: To understand and implement Document Type Definition (DTD) in XML and demonstrate its types (Internal and External DTD).

System Requirements –

Operating System: Windows 10 or above

RAM: Minimum 4 GB

Software/s Needed for Experiment–Web Browser, Text Editor, VSCode

Experiment Objectives:

- To understand the concept of DTD in XML.
- To learn the purpose and structure of DTD.
- To differentiate between Internal and External DTD.
- To validate XML documents using DTD.

Experiment Outcomes:

- Understand how DTD defines the structure of XML documents.
- Implement Internal DTD within an XML document.
- Implement External DTD using a separate .dtd file.
- Validate XML documents using defined rules.

Theory:

Document Type Definition (DTD) is used to define the structure and legal elements and attributes of an XML document. It specifies the rules that an XML document must follow. DTD ensures that XML data is structured and valid.

Types of DTD:

1. Internal DTD – Defined inside the XML document.
2. External DTD – Defined in a separate .dtd file and linked to the XML document.

Basic DTD Declarations:

- <!ELEMENT> – Defines elements
- <!ATTLIST> – Defines attributes
- <!ENTITY> – Defines entities

Program 1: Internal DTD Example

XML Code:

```
<?xml version="1.0"?>
<!DOCTYPE student [
  <!ELEMENT student (name, rollno, branch)>
  <!ELEMENT name (#PCDATA)>
  <!ELEMENT rollno (#PCDATA)>
```

```
<!ELEMENT branch (#PCDATA)>
]>
<student>
<name>Amruta</name>
<rollno>101</rollno>
<branch>Computer Engineering</branch>
</student>
```

Program 2: External DTD Example

Step 1: Create student.dtd file

```
<!ELEMENT student (name, rollno, branch)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT rollno (#PCDATA)>
<!ELEMENT branch (#PCDATA)>
```

Step 2: Create XML file and link external DTD

```
<?xml version="1.0"?>
<!DOCTYPE student SYSTEM "student.dtd">
<student>
<name>Amruta</name>
<rollno>102</rollno>
<branch>Information Technology</branch>
</student>
```

Task for Students: Create an XML document for a **Student Information System** containing the following elements:

<student>, <name>, <rollno>, <branch>, <email>

Task Requirements:

- Create an XML Document
- Implement Internal DTD
- Implement External DTD
- Validation Testing

Submission Requirement:

- XML file with Internal DTD
- External DTD file (.dtd)
- XML file linked with External DTD
- Screenshot of successful validation

- Screenshot showing validation error

Observations:

The XML document was successfully validated using both Internal and External DTD. Internal DTD was declared within the XML file, while External DTD was stored separately and linked using SYSTEM keyword.

Conclusion:

This experiment demonstrated the concept of DTD and its types. Students understood how DTD defines the structure of XML documents and how validation ensures proper data organization. Both Internal and External DTD implementations were successfully performed.

Expected Oral Questions:

1. What is DTD?
2. Why is DTD used in XML?
3. What is the difference between Internal and External DTD?
4. What are basic DTD declarations?

FAQs in Interview

1. What are the types of DTD?
2. What is `<!ELEMENT>` in DTD?
3. What is `<!ATTLIST>` in DTD?
4. What is `<!ENTITY>` in DTD?
5. What is `#PCDATA`?
6. What is the use of `<!DOCTYPE>` declaration?
7. What is the purpose of the SYSTEM keyword?
8. How do you validate an XML document using DTD?
9. What are the limitations of DTD?
10. What is the difference between DTD and XML Schema?
11. Can DTD define attributes?
12. Can DTD define default attribute values?
13. Is DTD case sensitive?
14. Can multiple XML files use the same External DTD?
15. What happens if XML does not follow DTD rules?